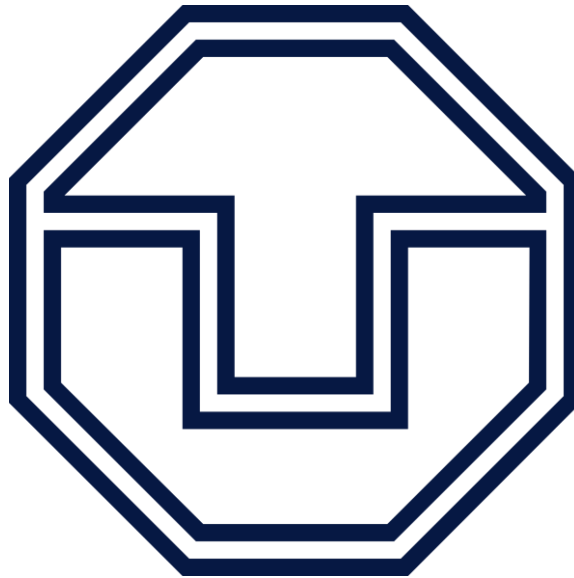


Hydrochemistry Laboratory Report

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Course of Studies : MSc. Hydro Science and Engineering
Module : MHSE08 - Hydrochemistry
Experiment Name : Chemical De-acidification using
calcium carbonate and semi burnt dolomite
Experiment Date : 5 January 2026



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1. Aim of the experiment

The aim of the experiment is to evaluate the suitability and efficiency of 4 commercial materials (HYDROLIT-Ca, HYDRO-Carbonate, AKDOLIT-GRAN, and MAGNO-DOL) to deacidify water. This experiment imitates the drinking water generation at water treatment plants. The experiment is performed by passing water through 4 separate column reactors filled with different filter materials mentioned earlier.

Since dolomite-based materials have faster kinetic reaction with CO_2 , it is hypothesized that AKDOLIT-GRAN in Reactor 3 and MAGNO-DOL in Reactor 4 will perform the most efficiently. However, MAGNO-DOL will perform slightly less efficient due to its splinter shape. The HYDROLIT-Ca and HYDRO-Carbonate materials in Reactors 1 and 2 will perform hypothetically the least efficient because both materials are calcium carbonate-based materials which are slower in neutralizing CO_2 . However, due to its 10% MgCO_3 and its spherical shape, HYDROLIT-Ca will perform slightly more efficient compared to HYDRO-Carbonate.

2. Theoretical background

Calco-carbonic equilibrium

Calco-carbonic equilibrium is a condition where water neither has excess of calcium carbonate nor lacking in it. Water is considered in Calco-carbonic equilibrium when it has the balance of Ca^{2+} and carbonic acids (CO_2 , HCO_3^- and CO_3^{2-}).

Total Hardness

Total hardness is the sum of dissolved alkaline earth metals (mainly Ca^{2+} and Mg^{2+}) in water. This term can be divided into carbonate and non-carbonate hardness.

Carbonate Hardness (Temporary hardness)

Carbonate Hardness or temporary hardness is the concentration of alkaline earth metal ions, which corresponds to the available concentration of dissolved bicarbonate (HCO_3^-) ions and can be removed by heating.

Permanent hardness

Non-carbonate hardness or permanent hardness is hardness caused by calcium and magnesium salts other than carbonates and cannot be removed by heating

Acid neutralising capacity

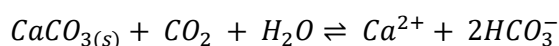
Acid neutralising capacity is the amount of acid consumed during a titration to a certain pH. In this experiment, we determine this to the pH 4.3, therefore it is called $\text{ANC}_{4.3}$.

Base neutralising capacity

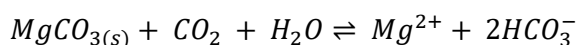
Base neutralising capacity is the amount of base consumed during a titration to a certain pH. In this experiment, we determine this to the pH 8.2, therefore it is called $\text{BNC}_{8.2}$.

Chemical equations responsible for de-acidification

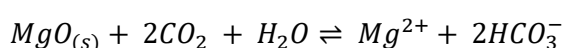
- Reaction with Calcium carbonate (Happens in all reactors)



- Reaction with Magnesium carbonate (Happens in Reactor 1)



- Reaction with MgO (Happens in Reactor 3 and 4)



3. Experimental

The experiment was conducted by following the procedure explained in the Experimental part of the “Deacidification of water – MHSE08” script. During the experiment, there was no deviation from the script procedure and no special observation.

4. Results

Raw data and calculations based on personal experiment

The reactor that our group worked on is the Reactor 4, which uses MAGNO-DOL as the filter material. The determined raw data (titrated volumes) is presented in Table 1

Table 1. Determined raw data (titrated volumes)

Water volume	Titration	Titration volume
100 mL sample water	0.1 M HCl	11.5 mL
100 mL sample water	0.1 M HCl	2.5 mL
50 mL sample water	0.01 M EDTA	0.9 mL
25 mL sample + 25 mL pure water	0.01 M EDTA	14.7 mL

Based on this titrated volume data, we performed the calculations to obtain Acid Neutralising Capacity to pH 4.3 ($ANC_{4.3}$), Acid Neutralising Capacity to pH 8.2 ($ANC_{8.2}$), calcium ions concentration ($c(Ca^{2+})$), and magnesium ions concentration ($c(Mg^{2+})$).

1. Acid Neutralising Capacity to pH 4.3 ($ANC_{4.3}$)

$$ANC_{4.3} = \frac{V_T \times C_T}{V_S} \times 1000 = \frac{11.5 \text{ mL} \times 0.1 \text{ mol/L}}{100 \text{ mL}} \times 1000 = 11.5 \text{ mmol/L}$$

2. Acid Neutralising Capacity to pH 8.2 ($ANC_{8.2}$)

$$ANC_{8.2} = \frac{V_T \times C_T}{V_S} \times 1000 = \frac{2.5 \text{ mL} \times 0.1 \text{ mol/L}}{100 \text{ mL}} \times 1000 = 2.5 \text{ mmol/L}$$

3. Calcium ions concentration ($c(Ca^{2+})$)

$$c(Ca^{2+}) = \frac{V_T \times C_T}{V_S} \times 1000 = \frac{0.9 \text{ mL} \times 0.01 \text{ mol/L}}{50 \text{ mL}} \times 1000 = 0.18 \text{ mmol/L}$$

4. Magnesium ions concentration ($c(Mg^{2+})$)

$$c(Ca^{2+} + Mg^{2+}) = \frac{V_T \times C_T}{V_S} \times 1000 = \frac{14.7 \text{ mL} \times 0.01 \frac{\text{mol}}{\text{L}}}{25 \text{ mL}} \times 1000 = 5.88 \text{ mmol/L}$$

$$c(Mg^{2+}) = c(Ca^{2+} + Mg^{2+}) - c(Ca^{2+}) = 5.88 - 0.18 = 5.7 \text{ mmol/L}$$

Final data for all reactors and raw water

After each group performed the experiment, data from all reactors and raw water is collected and summarized in Table 2. However, due to certain circumstances during the experiment, the group that responsible for AKDOLIT-GRAN evaluation could not determine the $ANC_{8.2}$. Therefore, it is not possible to determine the concentration of CO_3^{2-} , CO_2 , and HCO_3^- . As the result, it is also not possible to determine the $c(CO_2)_{eq}$ and pH_{eq} . Without these data, we cannot determine whether the water that has been treated with AKDOLIT-GRAN is calcite-dissolving, calcite-saturated, or calcite-precipitating.

Table 2. Summary of the experimental results

Parameter	Raw water	HYDROLIT-Ca	HYDRO-Carbonate	AKDOLIT-GRAN	MAGNO-DOL
pH	5.1	7.7	6.8	8.7	9.6
T [°C]	19.9	20.2	19.5	19.9	19.6
K _{25°C} [μS/cm]	124	685	613	1060	771
ANC _{4.3} [mmol/L]	0.925	7.5	6.4	13.3	11.5
ANC _{8.2} [mmol/L]	-	-	-	-	2.5
BNC _{8.2} [mmol/L]	8.7	1.2	2.7	-	-
c(Ca ²⁺) [mmol/L]	0.46	1.96	3.2	0.7	0.18
c(Mg ²⁺) [mmol/L]	0.36	2	0.0	2.79	5.7

Tillman's and Langelier's Equation Calculation

Before the Tillman's and Langelier's Equation can be used to determine if the water is calcite-dissolving, calcite-saturated or calcite-precipitating, we need to evaluate several concentrations. These concentrations determination can be performed by using the specific formula shown in Table 3 and Table 4.

Table 3. Necessary concentrations for water sample with pH < 8.2

Parameter	Formula	Raw water	HYDROLIT-Ca	HYDRO-Carbonate
c(HCO ₃ ⁻) [mol/L]	$c(HCO_3^-) = ANC_{4.3}$	9.25E-04	7.50E-03	6.40E-03
c(CO ₂) [mol/L]	$c(CO_2) = BNC_{8.2}$	8.70E-03	1.20E-03	2.70E-03

Table 4. Necessary concentrations for water sample with pH > 8.2

Parameter	Formula	AKDOLIT-GRAN	MAGNO-DOL
c(HCO ₃ ⁻) [mol/L]	$c(HCO_3^-) = ANC_{4.3} - 2c(CO_3^{2-}) - c(OH^-)$	-	6.54E-03
c(CO ₂) [mol/L]	$c(CO_2) = \frac{f_{a1}}{K_{a1}} \times a(H^+) \times c(HCO_3^-)$	-	4.29E-06
c(CO ₃ ²⁻) [mol/L]	$c(CO_3^{2-}) = ANC_{8.2} - c(OH^-)$	-	2.46E-03
c(OH ⁻) [mol/L]	$c(OH^-) = 10^{-pOH}$	5.01E-06	3.98E-05
c(H ⁺) [mol/L]	$c(H^+) = 10^{-pH}$	2.00E-09	2.51E-10

Additionally, the activity coefficient is needed for the Tillman's and Langelier's Equation. It is shown in Table 5.

Table 5. Activity coefficient determination

Parameter	Formula	Raw water	HYDROLIT-Ca	HYDRO-Carbonate	AKDOLIT-GRAN	MAGNO-DOL
Ionic Strength (I) [mol/L]	$I = \frac{\kappa_{25^\circ C} [\mu S/cm]}{62000}$	2.00E-03	1.10E-02	9.89E-03	1.71E-02	1.24E-02
Activity coefficient (γ ₁)	$\log \gamma_1 = -0.5 \frac{\sqrt{I}}{1 + 1.4\sqrt{I}}$	0.95	0.90	0.90	0.88	0.89

After all the data needed is determined, Tillman's and Langelier's Equation can be performed as shown in Table 6 and Table 7, respectively.

- Tillman's Equation

$$c(\text{CO}_2)_{eq} = \frac{K_T}{f_T} \times c^2(\text{HCO}_3^-) \times c(\text{Ca}^{2+})$$

with $K_T = \frac{K_{a2}}{K_{sp} \times K_{a1}}$

$$f_T = \frac{1}{\gamma_1^6}$$

$$K_{sp} = 3.926 \times 10^{-9} \frac{\text{mol}^2}{\text{L}^2}$$

$$K_{a1} = 3.428 \times 10^{-7} \frac{\text{mol}}{\text{L}}$$

$$K_{a2} = 3.251 \times 10^{-11} \frac{\text{mol}}{\text{L}}$$

Table 6. Tillman's Equation Calculation

Parameter	Raw water	HYDROLIT-Ca	HYDRO-Carbonate	AKDOLIT-GRAN	MAGNO-DOL
Tillman's Constant (K_T) [L ² mol ²]	24156	24156	24156	24156	24156
Tillman's Correction Factor (f_T)	1.34	1.88	1.83	2.15	1.95
$c(\text{CO}_2)_{\text{meas}}$ [mol/L]	8.70E-03	1.20E-03	2.70E-03	-	4.29E-06
$c(\text{CO}_2)_{eq}$ [mol/L]	7.11E-06	1.41E-03	1.73E-03	-	9.55E-05
Condition	Calcite-dissolving	Calcite-precipitating	Calcite-dissolving	-	Calcite-precipitating

- Langelier's Equation

$$pH_{eq} = -\log K_{La} - \log f_{La} - \log c(\text{HCO}_3^-) - \log c(\text{Ca}^{2+})$$

with $f_{La} = \gamma_1^5$

$$K_{La} = \frac{K_{a2}}{K_{sp}}$$

$$K_{sp} = 3.926 \times 10^{-9} \frac{\text{mol}^2}{\text{L}^2}$$

$$K_{a2} = 3.251 \times 10^{-11} \frac{\text{mol}}{\text{L}}$$

Table 7. Langelier's Equation Calculation

Parameter	Raw water	HYDROLIT-Ca	HYDRO-Carbonate	AKDOLIT-GRAN	MAGNO-DOL
Sum activity coefficient (f_{La})	0.78	0.59	0.61	0.53	0.57
Langelier's Constant (K_{La}) [L/mol]	8.28E-03	8.28E-03	8.28E-03	8.28E-03	8.28E-03
pH_{meas}	5.1	7.7	6.8	8.7	9.6
pH_{eq}	8.56	7.14	6.99	-	8.25
Saturation Index (SI)	-3.46	0.56	-0.19	-	1.35
Condition	Calcite-dissolving	Calcite-precipitating	Calcite-dissolving	-	Calcite-precipitating

5. Discussion

The initial raw water has a -3.46 Saturation Index, which means this water is calcite-dissolving and highly corrosive. Therefore, it is necessary to perform deacidification for this water. In this part, the efficiency of each material, the suitability for drinking water generation and its compliance with the German drinking water standards is explained.

Based on the results that are acquired on the previous section, HYDROLIT-Ca is considered the most effective material to deacidify raw water. The saturation index (SI) of 0.56 is above the desired Calco-carbonic equilibrium which means it will precipitate calcite. Although the gap to the desired Calco-carbonic equilibrium is closer for the water treated by HYDRO-Carbonate, the negative SI value means the HYDRO-Carbonate treatment failed to deacidify the water and will dissolve calcite and therefore considered corrosive. However, this can be fixed by adding extra contact time of the water with the material.

MAGNO-DOL succeed to deacidify the water. However, the pH 9.6 is higher than the maximum pH allowed by the German Drinking Water Ordinance, which stated that the pH of drinking water should be between pH 6.5 and pH 9.5. And like HYDROLIT-Ca treated water, this water will precipitate calcite even more which possibly also exceed the regulation of the maximum Calcite Dissolution Capacity (5mg/L). In contrast with the HYDRO-Carbonate, MAGNO-DOL still can be used for deacidification purpose by reducing the contact time with the water, so the pH will not reach 9.5. Finally, although specific calculation for AKDOLIT-GRAN treated water is impossible due to missing $ANC_{8.2}$, the pH of 8.7 strongly indicates an oversaturated condition (Calcite-precipitating) like MAGNO-DOL, as it significantly exceeds the equilibrium pH.

In summary, HYDROLIT-Ca is considered the most effective material to deacidify raw water for this specific contact time, although it still precipitates some calcite. HYDRO-Carbonate also can be utilized to deacidify water by increasing the contact time and MAGNO-DOL, on the other can be utilized as deacidification material by reducing the contact time.

6. References

- [1] Lab Script: Institute of Water Chemistry, TU Dresden. Lab Class MHSE 08 - Hydrochemistry, Experiment: Chemical De-acidification using calcium carbonate and semi-burnt dolomite.
- [2] Trinkwasserverordnung: Verordnung über die Qualität von Wasser für den menschlichen Gebrauch (Trinkwasserverordnung) [Ordinance on the Quality of Water Intended for Human Consumption]. Germany. Parameters for chemical quality: Calcite dissolution capacity & Hydrogen ion concentrations